

Tax & Labour-Code Compliance Tracker

ICAI AICA Level 2 Capstone Project — Project Summary

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1. Objective

Chartered accountancy teams track compliance due dates (GST, Income Tax, TDS, and labour-code obligations such as PF/ESI/LWF) manually, with reminders depending on someone remembering to chase them. This project builds an end-to-end, mostly automated system that: (1) maintains every due date in one extensible tracker; (2) escalates reminders automatically as a deadline approaches and passes; (3) gives each department and the HOD a live dashboard; (4) produces a month-end PDF report; and (5) reconciles actual tax payments against the trial balance and flags variances.

2. Architecture

A Google Sheet (imported from Compliance_Master.xlsx) is the single live source of truth for multi-user data entry. Editing access (Amount Paid, Payment Date, Task Complete) is controlled by Google Sheets' own Share/Editor permissions, restricted to specific invited users. n8n runs on a daily schedule, reads and writes that same Sheet, computes which escalation stage (T-2 / T-1 / due-or-overdue-daily) each unpaid task is in, and sends the matching email via Gmail, then logs the reminder back. The Streamlit dashboard reads the Sheet's "Task Instances" and "Trial Balance" tabs live via Google's "Publish to web" CSV export — a public read-only link, kept separate from the edit permissions above, needing no login to view. The dashboard can also generate the month-end PDF report and the trial-balance Variance Report on demand, as downloads, from the same live data.

3. Deliverables

Deliverable	File(s)	What it does
Master tracker	Compliance_Master.xlsx	6 sheets: ReadMe, Task Templates (extensible catalogue), Users, Task Instance
Reminder automation	n8n workflow: 'Compliance Reminder Escalation Workflow' (see the workflow export)	Derives escalation workflow from the tracker, computes escalation stage, routes and s
Dashboard	dashboard.py + requirements.txt	Streamlit app reading live from the shared Google Sheet: Due This Month / Comp
Month-end report	generate_monthly_report.py (standalone) + dashboard.py	Generates a PDF dashboard by department, listing due date, amount due/paid, pay
Variance check	generate_variance_report.py (standalone) + dashboard.py	Compares actual to dashboard (and by code, from the tracker) against the Trial Balance
Presentation	Compliance Tracker Capstone Walkthrough (Arwa.tts)	Walkthrough (Arwa.tts) speaker notes, used as the script for the video walkthrough

4. Escalation logic

- **Stage 1 (T-2 days):** email to the assigned user only.
- **Stage 2 (T-1 day):** if still unpaid, same reminder repeats with the supervisor cc'd.
- **Stage 3 (due date onward):** if still unpaid, the reminder goes to user, supervisor and HOD, and repeats every day until Amount Paid is filled in AND Task Complete = Y.

Frequency of the whole cycle follows the task's own periodicity automatically — monthly tasks (TDS, GSTR-1/3B) repeat every month, quarterly tasks (advance tax, TDS/TCS returns) four times a year, annual tasks (tax audit, ITR, GSTR-9C) once.

5. Data privacy note

Since this workbook and workflow are published as part of the capstone submission, no third-party names or email addresses are used anywhere. All User / Supervisor / HOD roles use only the account holder's own two email addresses (riteshgarodia@yahoo.com as User, garodia1.ritesh@gmail.com as Supervisor/HOD) as a stand-in for a real multi-person team.

6. Known simplifications for this demo

- Dashboard uses a View toggle (User/HOD) instead of email-based login, to keep the demo/recording simple. A live deployment would authenticate by email and route each person to their own department automatically.
- Dashboard is run locally for this demo (streamlit run dashboard.py), so it is only reachable on the presenter's own machine. A live deployment would host it on Streamlit Community Cloud (or a similar host) so it runs continuously at one shared URL every user and the HOD can open from their own browser.
- A handful of labour-code due dates (Professional Tax, LWF) are state-specific and are marked as assumptions in the Task Templates sheet — verify against local rules before production use.